AP Calc Lesson 2.4 Homework

Answer Key

1. $f'(x) = 3x^2 + \frac{2}{\sqrt{x}} - \frac{6}{x^2}$

2. a. $f'(x) = 3x^2 + 12x - 15$

   b. $3(x+5)(x-1) = 0$

   $x = -5 \text{ and } x = 1$

3. a. $f(3) = 4(3)^3 + 2(3) + 5g(3) = 108 + 6 + 6(-4) = 114 - 24 = 90$

   b. $f'(x) = 12x^2 + 2 + 6g'(x)$

   c. $y - 90 = 106(x - 3)$ or equivalent

   Note: $f'(3) = 12(3)^2 + 2 + 6g'(3) = 108 + 2 + 6\left(-\frac{2}{3}\right) = 110 - 4 = 106$

4. a. $\frac{x(9) - x(0)}{9 - 0} = \frac{181 - 1}{9} = 20$

   b. $x'(t) = 12t^{1/2} - 4$

   c. $12t^{1/2} - 4 = 20 \Rightarrow t^{1/2} = 2 \Rightarrow t = 4$

5. a. $f'(x) = \frac{-4}{x^{1/3}} + b$

   b. $f'(8) = -2 + b$

6. $a = 2, b = 0.5$, and $c = 6$.

   $f(2) = 2a - 4 = 0 \rightarrow a = 2$ and $g(x) = bx^2 - 4x + c$

   $f'(x) = 2 - 2x$ and $g'(x) = 2bx - 4$, so $2 - 2x = 2bx - 4$ at $x = 2 \rightarrow 2 - 4 = 4b - 4$, so $b = 0.5$.

   $g(2) = 0.5(2)^2 - 4(2) + c = 0 \rightarrow 2 - 8 + c = 0 \rightarrow c = 6$

7. $g'(2) = 27$

8. The linear term of f becomes the constant term/ y -intercept of f, so $a^2 = 16$ and $a = \pm 4$. The two possible values of a give two possible options for all the other coefficients so either $f(x) = -4x^4 - 7x^3 - 4x^2 + 16x - 4$ or $f(x) = 4x^4 + x^3 + 4x^2 + 16x + 4$.

9. C

10. A